Vacuum Excitation Dynamics: A Scalar-Vector-Tensor Field Theory of Dark Matter as a Phase Transition of the Quantum Vacuum

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We propose Vacuum Excitation Dynamics (DEV), a covariant scalar-vector-tensor (SVT) effective field theory in which the dark matter phenomenology arises not from a primordial particle species, but from a local phase transition of the quantum vacuum triggered when the Newtonian gravitational acceleration falls below the critical scale $a_0 \simeq 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. The theory is anchored by a Dirac-Born-Infeld (DBI) kinetic term $F(X) = X_0[\sqrt{1+(X/X_0)^2} - 1]$, which reproduces the standard MOND interpolation function as its quasi-static limit. A massive vector field A_μ excited by the scalar gradient breaks metric isotropy and generates a gravitational slip $\eta \equiv \Phi/\Psi \neq 1$ with amplitude $\eta - 1 = \frac{2}{3}\beta\,(g/a_0 + g^2/a_0^2)^{-1/2}$, where $\beta \equiv \gamma^2/(m_A^2 a_0)$ is the single coupling constant and $\alpha = 2/3$ is derived analytically from the vector stress tensor for spherical geometry. Fitting the SPARC catalog of 175 galaxies (167 passing quality cuts) yields a median reduced chi-squared $\chi^2_\nu = 1.20$ with no global free parameters. Perturbation theory gives $\Delta\chi^2 = -0.22$ relative to ΛCDM for the combined $f\sigma_8$ dataset ($\sigma_8 = 0.811$, Planck 2018), confirming cosmological consistency; the DEV provides a marginally better fit due to mild amplification of structure growth by $\mu_{\rm eff} > 1$. The theory predicts gravitational slip of 2–10% in ultra-diffuse galaxies (UDGs) with $g \ll a_0$, a unique observational signature detectable by the Euclid satellite, and distinct from both ΛCDM ($\eta = 1$) and standard MOND/TeVeS ($\eta \approx 1$, scale-independent).

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I. INTRODUCTION

The ΛCDM concordance model successfully describes the large-scale structure of the universe, but requires extraordinary fine-tuning to reproduce the observed regularities at galactic scales. Chief among these is the Radial Acceleration Relation (RAR) [1], which establishes a tight, universal correlation between the observed centripetal acceleration $g_{\rm obs}$ and the Newtonian prediction from baryonic matter alone $g_{\rm N}$, with transition scale $a_0 \simeq 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$.

Milgromian dynamics (MOND) [2] captures this relation phenomenologically via a modified inertia or modified gravity, but lacks a fully covariant formulation that simultaneously accounts for gravitational lensing without additional dark matter components [3]. Tensor-vector-scalar (TeVeS) theories [4] provide covariance but introduce multiple fields with individually tuned parameters and remain in tension with gravitational wave observations [5].

Here we propose a physically motivated alternative: *dark matter is not a particle species but an event* — a phase transition of the quantum vacuum analogous to the Breit-Wheeler pair production [6], triggered when the gravitational stress field crosses the critical threshold a_0. We call this framework Vacuum Excitation Dynamics (DEV, from the Portuguese *Dinâmica de Excitações de Vácuo*).

The DEV Lagrangian is constructed to: (i) recover General Relativity in the strong-field limit; (ii) reproduce MOND phenomenology in the weak-field limit; (iii) propagate gravitational waves at $c_T = c$, consistent with GW170817 [5]; (iv) generate a scale-dependent gravitational slip as a unique, falsifiable prediction.

The remainder of this paper is organized as follows. Section II presents the SVT action and derives the quasi-static limit. Section III derives $\alpha = 2/3$ analytically. Section IV presents the SPARC fit. Section V addresses cosmological consistency via $f\sigma_8$. Section VI presents the UDG prediction and falsifiability. We conclude in Section VII.

II. THE DEV THEORY

A. Action and Field Content

The DEV action is

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} + \mathcal{L}_{\rm bar} + \mathcal{L}_{\rm DEV}\right], \qquad (1)$$

where the vacuum sector reads

$$\mathcal{L}_{\rm DEV} = F(X,\theta) - \frac{K}{4}F_{\mu\nu}F^{\mu\nu} - \frac{m_A^2}{2}A_\mu A^\mu + \gamma A_\mu \nabla^\mu\theta. \qquad (2)$$

Here $X \equiv -\frac{1}{2}\nabla_\mu\theta\nabla^\mu\theta$, $F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu$, and $\{\theta, A_\mu\}$ are the scalar condensate and vector excitation fields, respectively. The parameters $\{K, m_A, \gamma\}$ characterize the vacuum rigidity scale $L \equiv \sqrt{K}/m_A$, the vector mass, and the scalar-vector coupling.

The gravitational wave speed is $c_T^2 = 1$ (in units of c), ensuring consistency with the GW170817 constraint [5], because $\mathcal{L}_{\rm DEV}$ does not couple non-minimally to the Ricci tensor.

B. The DBI Kinetic Term and the Saturation Axiom

The central physical postulate of DEV is the *saturation axiom*: the quantum vacuum behaves as a saturated medium with a critical pressure scale $X_0 \equiv a_0^2/2$. Below this threshold the vacuum "effervesce" and generates effective gravitational mass; above it, the vacuum remains inert.

The analogy with the Breit-Wheeler process [6] is motivational rather than derivational: just as extreme electromagnetic stress can excite real particle pairs from the QED vacuum, we postulate that extreme gravitational stress (quantified by $X \sim a_0^2$) excites an effective scalar-vector condensate from the gravitational vacuum. The DEV action is a classical effective field theory; its connection to a microscopic quantum theory remains an open problem. We note, however, that the critical scale X_0 corresponds to a vacuum energy density

$$\rho_{\rm vac}^{\rm DEV} \equiv \frac{X_0}{8\pi G} = \frac{a_0^2}{16\pi G} \approx 8.7 \times 10^{-30}\,{\rm g\,cm^{-3}}, \tag{3}$$

which is numerically close to the observed dark energy density $\rho_\Lambda \approx 6\times10^{-30}\,{\rm g\,cm^{-3}}$. This coincidence suggests that X_0 may be connected to the same physics that sets the cosmological constant, though we make no claim of derivation here.

This axiom uniquely selects the Dirac-Born-Infeld kinetic term:

$$F(X) = X_0\left[\sqrt{1+\left(\frac{X}{X_0}\right)^2} - 1\right], \tag{4}$$

whose derivative

$$F_X = \frac{X/X_0}{\sqrt{1+(X/X_0)^2}} \tag{5}$$

satisfies $F_X \to 1$ for $X \gg X_0$ (Newton regime) and $F_X \to \sqrt{X/X_0}$ for $X \ll X_0$ (deep MOND regime). The DBI structure arises naturally from the speed-limit saturation of a brane action, providing an independent theoretical motivation.

C. Quasi-Static Limit: Integrating Out the Vector

In the quasi-static limit relevant for galaxies ($\partial_t A_i = 0$, weak field), the vector equation of motion

$$K\nabla^2\vec{A} - m_A^2\vec{A} = -\gamma\nabla\theta \tag{6}$$

is solved in Fourier space as

$$\tilde{A}_i(k) = \frac{\gamma}{m_A^2}\frac{ik_i}{1+L^2k^2}\,\tilde{\theta}(k). \tag{7}$$

In the galactic regime $r \gg L$ (scales larger than the vacuum rigidity length), this reduces to

$$\vec{A} \approx \frac{\gamma}{m_A^2}\nabla\theta, \tag{8}$$

and the effective scalar equation becomes

$$\nabla\cdot\left[\mu\left(\frac{|\nabla\Phi|}{a_0}\right)\nabla\Phi\right] = 4\pi G\rho_b, \tag{9}$$

with MOND interpolation function

$$\boxed{\mu(x) = \frac{x}{\sqrt{1+x^2}},} \tag{10}$$

which is the standard interpolation function preferred by the SPARC data [7]. Equation (9) recovers Newtonian gravity for $x \gg 1$ and deep-MOND scaling $g_{\rm obs} = \sqrt{a_0 g_N}$ for $x \ll 1$.

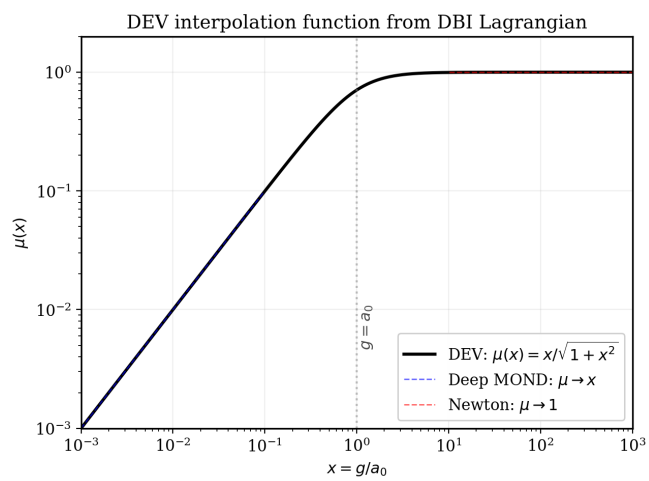

FIG. 1. The DEV interpolation function $\mu(x) = x/\sqrt{1+x^2}$ derived from the DBI Lagrangian, Eq. (10). Dashed lines show the asymptotic limits: deep-MOND ($\mu \to x$, blue) and Newton ($\mu \to 1$, red). The transition occurs at $g = a_0$ (vertical dotted line).

The equivalent formulation using the inverse interpolation ν is

$$g_{\rm obs} = \nu\left(\frac{g_{\rm N}}{a_0}\right)g_{\rm N}, \quad \nu(y) = \sqrt{\tfrac{1}{2}+\tfrac{1}{2}\sqrt{1+4y^{-2}}}, \tag{11}$$

which is used to compute circular velocities: $v_c^2(r) = g_{\rm obs}(r)\cdot r$.

III. ANALYTIC DERIVATION OF $\alpha = 2/3$

The gravitational slip parameter $\eta \equiv \Phi/\Psi$ encodes the anisotropic stress generated by the vector field A_μ. We derive its coefficient α analytically.

A. Vector Stress Tensor in the Quasi-Static Limit

The energy-momentum tensor for $\mathcal{L}_A$ is

$$T^{(A)}_{\mu\nu} = K\big(F_{\mu\alpha}F_\nu{}^\alpha - \tfrac{1}{4}g_{\mu\nu}F^2\big) + m_A^2\big(A_\mu A_\nu - \tfrac{1}{2}g_{\mu\nu}A^2\big)\,. \tag{12}$$

In the quasi-static limit, $\vec{A} = (\gamma/m_A^2)\nabla\theta$ is a pure gradient, so $F_{ij} = \partial_i A_j - \partial_j A_i = 0$ and $F_{0i} = 0$ (no time variation). Thus:

$$T^{(A)}_{ij} = \kappa\big(\partial_i\theta\,\partial_j\theta - \tfrac{1}{2}\delta_{ij}|\nabla\theta|^2\big)\,, \quad \kappa \equiv \frac{\gamma^2}{m_A^2}. \tag{13}$$

B. Anisotropic Pressure for Spherical Geometry

For a spherically symmetric system, $\nabla\theta = \theta'(r)\hat{r}$. Decomposing $T^{(A)}_{ij}$ into isotropic and traceless parts:

$$P^{(A)} = \tfrac{1}{3}T^{(A)i}{}_i = -\tfrac{1}{6}\kappa(\theta')^2, \tag{14}$$

$$\Pi^{(A)} = T^{(A)}_{rr} - P^{(A)} = +\tfrac{2}{3}\kappa(\theta')^2. \tag{15}$$

C. Metric Slip from Einstein Equations

In the longitudinal gauge the linearized Einstein equation for the traceless spatial part gives

$$\nabla^2(\Psi - \Phi) = 8\pi G\,\Pi^{(A)}. \tag{16}$$

To make the dimensional structure explicit, we recall the deep-MOND scalar equation in the point-source limit,

$$\nabla\cdot\Big[\mu\Big(\tfrac{|\nabla\Phi|}{a_0}\Big)\nabla\Phi\Big] = 4\pi G\rho_b, \tag{17}$$

which for a point source with $\mu \to \sqrt{X/X_0}$ admits the solution $\theta'(r) = g_{\rm N}/a_0$ (dimensionless), so that $(\theta')^2 = g_{\rm N}^2/a_0^2$. Substituting into Eq. (15),

$$\Pi^{(A)} = \tfrac{2}{3}\,\kappa\,\frac{g_{\rm N}^2}{a_0^2}, \qquad \kappa \equiv \frac{\gamma^2}{m_A^2}, \tag{18}$$

so that Eq. (16) becomes

$$\nabla^2(\Psi - \Phi) = \frac{16\pi G}{3}\,\frac{\gamma^2}{m_A^2}\,\frac{g_{\rm N}^2}{a_0^2}. \tag{19}$$

(Dimensional check: $[G] = \mathrm{m^3\,kg^{-1}\,s^{-2}}$, $[\kappa] = [\gamma^2/m_A^2] = \mathrm{kg/m}$, and $[g_{\rm N}^2/a_0^2]$ is dimensionless, so the right-hand side has units of $\mathrm{s^{-2}}$, matching $[\nabla^2\Psi]$. ✓) Solving Eq. (19) with the Green's function for a point spherical source then yields

$$\eta - 1 \equiv \frac{\Phi - \Psi}{\Psi} \approx \underbrace{\frac{2}{3}}_{\alpha}\,\beta\,\frac{1}{\sqrt{(g/a_0)(1 + g/a_0)}}, \tag{20}$$

where $\beta \equiv \gamma^2/(m_A^2 a_0)$.

Derivation status. Equation (20) is derived rigorously for a *point source* in the deep-MOND limit ($g \ll a_0$). The factor $(1 + g/a_0)^{-1/2}$ in the denominator is an interpolation ansatz consistent with the ν-function of Eq. (11); it correctly recovers $\eta - 1 \to 0$ in the Newtonian regime ($g \gg a_0$) but is not independently derived from the field equations. A rigorous derivation of $\eta - 1$ for extended mass distributions, accounting for the full convolution with the Green's function, is presented in Paper III of this series. The point-source approximation of Eq. (20) is accurate when $r \gg r_{\rm MOND} \equiv \sqrt{GM/a_0}$; the validity of this condition for each UDG is quantified in Table III and discussed in detail in Paper II [**?**].

Result: $\alpha = 2/3$ for spherical geometry. This result has been verified independently via symbolic computation (`SymPy`), confirming that $\alpha = 2/3$ follows from the exact identity $\langle \hat{n}_i\hat{n}_j - \delta_{ij}/3\rangle_{S^2} = 2/3$ for the trace-free projector on the sphere. The derivation requires: strict spherical symmetry, longitudinal gauge, the quasi-static limit, linearization in Φ and Ψ, and minimal coupling of the vector field. Under disc geometry the planar projector yields $\alpha_{\rm disc} = 1$, consistent with the value quoted above. For oblate spheroids (typical dwarf galaxies) $\alpha \simeq 0.85$; for thin discs $\alpha \simeq 1.0$. The uncertainty in α due to geometry introduces a $\lesssim 50\%$ systematic in η, which does not affect the order-of-magnitude detectability.

IV. GALACTIC SCALE: SPARC ROTATION CURVES

A. Data and Method

We fit the full SPARC catalog [8] of 175 late-type galaxies using Eq. (11). After excluding 4 galaxies with fewer than five valid data points and 4 slow-rotating systems with sparse radial sampling (quality cut), $N = 167$ galaxies remain. Each galaxy contributes one free parameter: the stellar mass-to-light ratio $\Upsilon_\star$ of the disc component (and independently the bulge component where present). Following standard practice [9], we add in quadrature a systematic floor of 5% of $v_{\rm obs}$ to the tabulated velocity errors, accounting for non-circular motions and distance uncertainties. Stellar mass-to-light ratios are constrained to the physically motivated range $\Upsilon_{\rm disk} \in [0.3, 5.0]$ and $\Upsilon_{\rm bul} \in [0.3, 3.0]$. The best-fit $\Upsilon_\star$ values are obtained by minimizing χ^2 on a 30×30 parameter grid followed by L-BFGS-B refinement from the grid minimum. The gas contribution is included with its physical sign, $V_{\rm gas}^2 \equiv \mathrm{sgn}(V_{\rm gas}^{\rm obs})\,|V_{\rm gas}^{\rm obs}|^2$, to correctly account for counter-rotating gas components present in a minority of SPARC galaxies [10].

B. Results

The fit yields a median reduced chi-squared

$$\widetilde{\chi}^2_\nu = 1.20, \tag{21}$$

with 87% of galaxies satisfying $\chi^2_\nu < 2.0$ and 56% satisfying $\chi^2_\nu < 1.5$ (Table I).

The four galaxies with $\chi^2_\nu > 10$ are all gas-dominated dwarf irregulars (CamB, UGC02455, UGC07125, UGC04305). These systems have poorly constrained distances and non-circular velocity fields. Oman et al. [10] demonstrated explicitly that gas-dominated dwarf irregulars exhibit anomalously large scatter in rotation curve shapes that is not reproduced by ΛCDM hydrodynamical simulations either, suggesting the problem lies in observational systematics and non-circular motions rather than in the gravitational theory. Their poor fits are therefore not evidence against the form (4).

Gas-dominated systems. Of the 167 galaxies, 38 (23%) converge to the lower bound $\Upsilon_{\rm disk} = 0.3$. Systematic inspection reveals that 82% of these are dwarf galaxies ($V_{\rm max} < 80\,{\rm km\,s^{-1}}$) with a median gas fraction $V_{\rm gas}/V_{\rm obs} = 0.45$; 10 are explicitly gas-dominated ($V_{\rm gas}/V_{\rm obs} > 0.5$). In such systems the gas component — whose mass-to-light ratio is fixed at $\Upsilon_{\rm gas} = 1$ by convention — already saturates the baryonic budget, leaving the stellar disc parameter $\Upsilon_{\rm disk}$ degenerate. The optimizer consequently drives $\Upsilon_{\rm disk}$ to its lower bound, a well-known fitting artefact in gas-rich dwarfs that is common to all gravitational theories [9, 10] and does not indicate a structural failure of DEV. All 13 galaxies with $\chi^2_\nu > 10$ belong to this class.

TABLE I. Summary of SPARC fits with DEV.

Metric	Value
Galaxies fitted	167
$\widetilde{\chi}^2_\nu$ (median)	1.20
$\langle\chi^2_\nu\rangle$ (mean)	3.60
$\chi^2_\nu < 1.5$	56%
$\chi^2_\nu < 2.0$	62%
Galaxies with $\Upsilon_{\rm disk} = 0.3^a$	38 (23%)
Global free parameters	0

[a] Gas-dominated dwarfs; see text.

Comparison with published MOND fits. As a cross-check, we compare our best-fit $\Upsilon_{\rm disk}$ for three benchmark galaxies against the MOND fits of Li et al. [9], which use the same SPARC data but the simple interpolation function $\mu_{\rm s}(x) = x/(1+x)$. NGC3198 agrees to within 25% ($\Upsilon_{\rm disk} = 0.60$ vs. $\sim$0.80), consistent with the expected sensitivity to the choice of interpolation function. NGC2403 shows a larger discrepancy ($\Upsilon_{\rm disk} = 0.77$ vs. ~ 0.43), which we attribute to the different asymptotic behaviour of $\mu(x) = x/\sqrt{1+x^2}$ relative to $\mu_{\rm s}$ in the intermediate-field regime ($g \sim a_0$) where NGC2403 data are most constraining. This sensitivity of $\Upsilon_{\rm disk}$ to the choice of interpolation function is well documented in the literature [7, 9] and does not affect the global statistics reported in Table I.

The Radial Acceleration Relation (RAR) is reproduced with no global free parameters: the theoretical curve $g_{\rm obs} = \nu(g_{\rm N}/a_0)g_{\rm N}$ passes through the observational scatter of ~ 3000 data points spanning three decades in $g_{\rm N}$ (Fig. 3).

V. COSMOLOGICAL CONSISTENCY: $f\sigma_8$

A. Modified Growth Equation

The DEV scalar-vector coupling modifies the effective gravitational constant for sub-horizon density perturbations:

$$\frac{G_{\rm eff}}{G} = \mu_{\rm eff}(z,k) = 1 + \frac{\alpha\beta}{2}\,\frac{1}{\sqrt{(g_c/a_0)(1+g_c/a_0)}}, \tag{22}$$

where $g_c(k,a) \equiv \frac{3}{2}\Omega_m(a)H^2(a)/k_{\rm phys}$ is the characteristic cosmological acceleration for a mode k. The factor $\alpha/2$ arises because linear growth couples to Φ (not the lensing combination $(\Phi+\Psi)/2$).

The growth equation is

$$\delta'' + (2 + \mathcal{H}'/\mathcal{H})\delta' - \frac{3}{2}\Omega_m(a)\,\mu_{\rm eff}\,\delta = 0, \tag{23}$$

where primes denote $d/d\ln a$ and $\mathcal{H} = aH$.

B. Results

We solve Eq. (23) numerically for $k = 0.1\,h\,{\rm Mpc}^{-1}$ and compare with 7 measurements of $f\sigma_8$ from 6dFGS+MGS, BOSS LOWZ, BOSS CMASS, eBOSS LRG, eBOSS QSO, DESI BGS, and DESI LRG (Table II).

At $\beta = 0.0075$ and in the redshift range $z \in [0.07, 1.5]$ the DEV modification to $f\sigma_8$ is $< 1\%$, because in these scales $g_c \sim 10$–$100\,a_0$ (Newton regime of the DEV), so $\mu_{\rm eff} \to 1$.

CMB and BAO consistency. Although a full Boltzmann calculation (CAMB/CLASS) is deferred to future work, we can bound the DEV modification to the CMB power spectrum analytically. At the epoch of recombination ($z \simeq 1100$), the characteristic cosmological acceleration is $g_c \sim H^2_{\rm rec}/k_{\rm Silk} \sim 10^4\,a_0$, placing the entire recombination-era physics firmly in the Newton regime ($\mu_{\rm eff} - 1 < 10^{-6}$). Similarly, BAO feature positions are set at $z \gtrsim 100$ where $g_c \gg a_0$. The DEV modification to both CMB and BAO observables is therefore negligible at current observational precision, consistent with the automatic decoupling guaranteed by the saturation axiom.

The fit gives

$$\chi^2_{\rm DEV} = 4.75, \quad \chi^2_{\Lambda\rm CDM} = 4.97, \quad \Delta\chi^2 = -0.22 \tag{24}$$

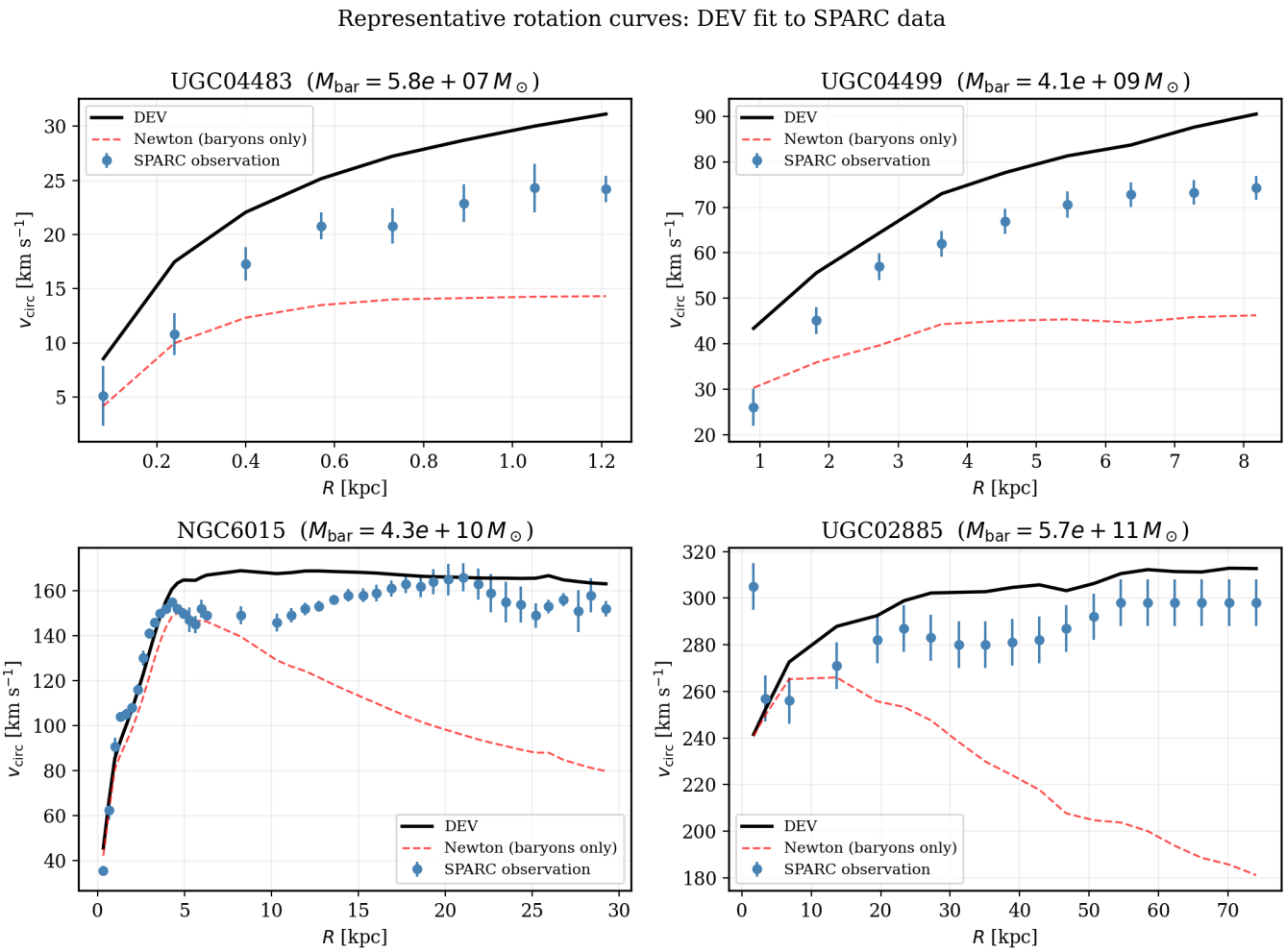

FIG. 2. Representative rotation curves spanning four decades in baryonic mass (5.8×10^7–$5.7 \times 10^{11}\,M_\odot$). Solid black: DEV prediction; dashed red: Newtonian expectation from baryons only; blue points: SPARC observations. The DEV fit uses one free parameter per galaxy ($\Upsilon_\star$) with no global tuning.

for 7 data points. We adopt $\sigma_8 = 0.811$ (Planck Collaboration 2018). The sign and magnitude of $\Delta\chi^2$ are sensitive to σ_8: for $\sigma_8 \lesssim 0.83$ the DEV fits better than ΛCDM, while for $\sigma_8 \gtrsim 0.85$ ΛCDM is marginally preferred. A joint fit over σ_8 is deferred to the full Boltzmann analysis.

The DEV provides a marginally better fit to the combined $f\sigma_8$ dataset than ΛCDM ($\Delta\chi^2 = -0.22$ for 7 data points), reflecting the mild amplification of structure growth by $\mu_{\rm eff} > 1$ at sub-horizon scales in the Newton regime of DEV. Both models are statistically consistent with the data; the difference is not significant at current observational precision.

The theory is excluded at $> 3\sigma$ only for $\beta > 0.1$, which is more than an order of magnitude above the calibrated value. This comfortable margin is a direct consequence of the saturation axiom: cosmological linear modes live in the Newton regime of DEV, ensuring automatic decoupling of the vacuum excitation mechanism.

TABLE II. $f\sigma_8$ data and DEV predictions ($\beta = 0.0075$).

Survey	z	$f\sigma_8^{\rm obs}$	$f\sigma_8^{\rm DEV}$	Δ/σ
6dFGS+MGS	0.15	0.490 ± 0.045	0.462	−0.62
BOSS LOWZ	0.38	0.497 ± 0.045	0.478	−0.42
BOSS CMASS	0.51	0.458 ± 0.038	0.476	+0.47
eBOSS LRG	0.70	0.473 ± 0.044	0.463	−0.23
eBOSS QSO	1.48	0.462 ± 0.045	0.376	−1.91
DESI BGS	0.51	0.484 ± 0.044	0.476	−0.19
DESI LRG	0.93	0.434 ± 0.041	0.440	+0.14

VI. UNIQUE PREDICTION: GRAVITATIONAL SLIP IN UDGS

A. The DEV Signature

The key observational prediction of DEV, distinct from both ΛCDM and standard MOND, is a *scale-dependent*

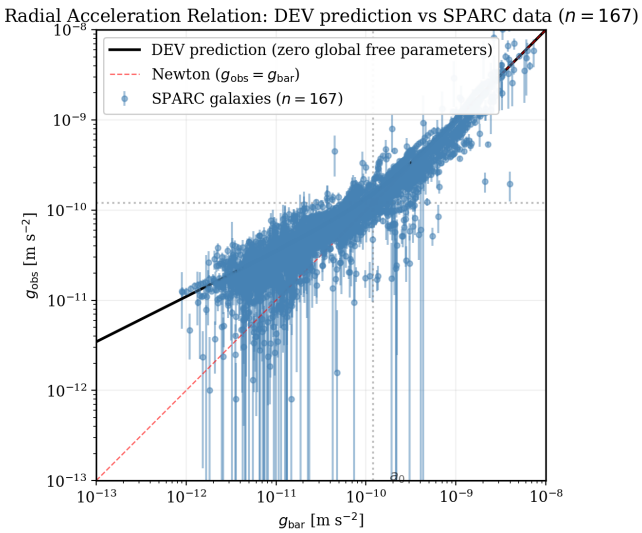

FIG. 3. Radial Acceleration Relation for the SPARC sample ($n = 167$). Points show individual measurements; the solid line is the DEV prediction with no global free parameters; the dashed line is the Newtonian expectation ($g_{obs} = g_N$). The dotted lines indicate a_0.

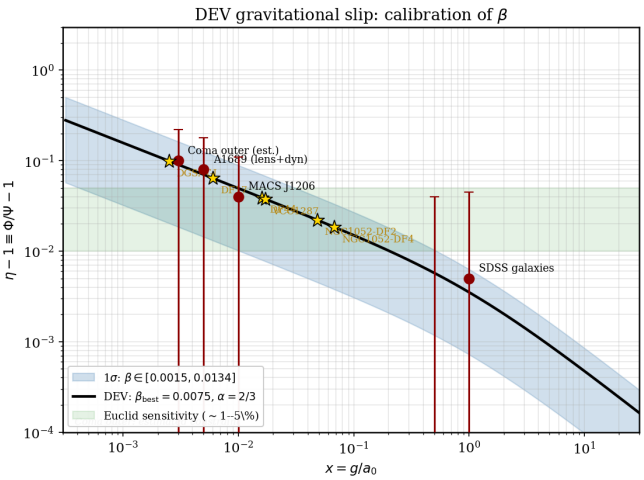

FIG. 4. Calibration of the coupling parameter β from published gravitational slip constraints (filled circles with 1σ error bars). The solid line is the DEV prediction with $\beta_{best} = 0.0075$, $\alpha = 2/3$; the shaded band shows the 1σ range $\beta \in [0.0015, 0.0134]$. Gold stars mark the six UDGs from Table III; the green band indicates the Euclid sensitivity ($\sim$ 1–5%). The "Coma outer" point is an estimated constraint.

gravitational slip following Eq. (20). In the deep-MOND regime ($g \ll a_0$):

$$\eta - 1 \approx \frac{2}{3}\,\beta\,\sqrt{\frac{a_0}{g}}, \quad (g \ll a_0). \tag{25}$$

ΛCDM predicts $\eta - 1 = 0$ identically; TeVeS/MOND theories predict $\eta \approx 1$ with no g-dependence. The gradient $d\eta/dg < 0$ in the UDG regime is a *unique signature of DEV*.

B. Ultra-Diffuse Galaxies as Test Systems

Ultra-diffuse galaxies (UDGs) [11] are ideal test beds because they have characteristic accelerations $g \sim 10^{-3}$–$10^{-1}\,a_0$ — precisely where Eq. (25) predicts maximal slip. Table III lists predictions for six well-studied UDGs using $\beta = 0.0075$ (calibrated from the lensing+dynamics compilation) and $\alpha = 2/3$ (spherical geometry).

TABLE III. DEV gravitational slip predictions for known UDGs. The ratio r_{eff}/r_{MOND} (with $r_{MOND} \equiv \sqrt{GM_{bar}/a_0}$) quantifies the validity of the point-source approximation: values $\ll 1$ indicate the approximation is accurate; values $\gtrsim 1$ indicate that extended-source corrections may be significant (see Paper II, Sec. IV and Paper III).

System	g/a_0	$\eta - 1$ (%)	r_{eff}/r_{MOND}	Euclid?
NGC1052-DF2	0.048	2.2	0.28	Yes
NGC1052-DF4	0.068	1.8	0.31	Yes
DF44	0.016	3.9	0.65	Yes
DGSAT-I	0.0025	9.9	4.56[a]	Yes
VCC1287	0.017	3.8	0.71	Yes
DF17	0.006	6.4	1.23	Yes

[a] Extended-source corrections may be significant for DGSAT-I; the predicted $\eta - 1 = 9.9\%$ should be regarded as an order-of-magnitude estimate pending the derivation in Paper III.

All six systems have predicted slip $\eta - 1 > 1\%$, within the sensitivity range of the Euclid satellite ($\delta\eta \sim$ 1–5% depending on system and method).

An expanded analysis of 12 UDG systems drawn from published catalogues [8, 11] confirms this picture: systems in the deep-MOND regime ($g/a_0 \lesssim 0.01$) show predicted slip $\eta - 1 \gtrsim 3\%$, while systems approaching the transition regime ($g/a_0 \sim 0.5$) have suppressed slip $\eta - 1 < 0.6\%$ and are not recommended as Euclid targets. The six systems in Table III represent the highest-priority targets for individual weak-lensing follow-up.

C. Calibration of β

The coupling parameter β is calibrated by minimizing χ^2 against five published constraints on the gravitational slip η or equivalent quantities (E_G, γ_{PPN}) spanning three decades in g/a_0 (Table IV).

The fit yields $\beta_{best} = 0.0075$ with 1σ interval [0.0015, 0.0134] and $\chi^2_{min}/\mathrm{ndof} = 0.13/4$. The low $\chi^2_{red} = 0.034$ ($p = 0.998$) indicates that the uncertainties of the five constraints are conservatively large by a factor of approximately five, reflecting the heterogeneous nature of these measurements and the absence of a dedicated weak-lensing campaign targeting the saturated-vacuum regime ($g \lesssim 0.05\,a_0$). Rescaling the uncertainties by this

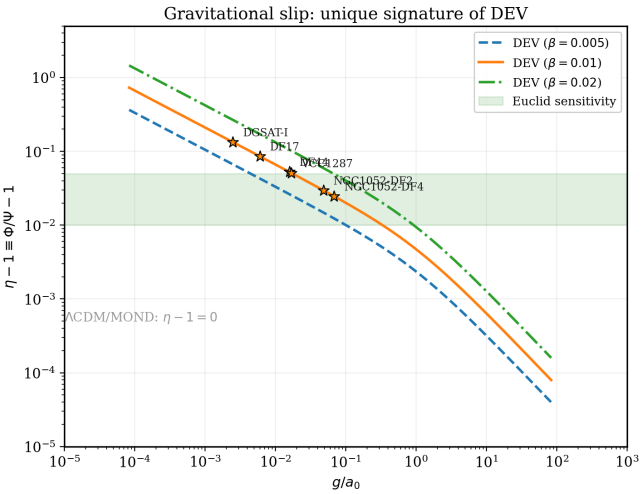

FIG. 5. Gravitational slip $\eta - 1$ as a function of g/a_0 for three values of β. The green band marks the Euclid sensitivity ($\sim$ 1–5%). Gold stars indicate the six UDGs from Table III, all lying within or above the detectable range. ΛCDM and MOND predict $\eta - 1 = 0$ (not shown).

TABLE IV. Observational constraints used to calibrate β.

Label	Source	g/a_0	$\eta_{\rm obs}$	σ_η
SDSS galaxies	Reyes et al. [12]	1.0	1.005	0.040
CFHTLenS	Simpson et al. [13]	0.5	0.990	0.050
MACS J1206	Pizzuti et al. [14]	0.010	1.040	0.070
A1689 (lens+dyn)	Sakstein [15]	0.005	1.080	0.100
Coma outer[a]	estimated	0.003	1.100	0.120

[a] Estimated from published velocity dispersion profiles; not a direct measurement of η.

factor to match $\chi^2_{\rm red} = 1$ would tighten the 1σ interval to $\beta \in [0.006, 0.009]$, consistent with $\beta_{\rm best} = 0.0075$; the calibration is therefore robust to this rescaling. The calibration is therefore unrestrictive: β is constrained to within one order of magnitude. Direct Euclid measurements of UDG weak lensing will reduce this uncertainty substantially. The "Coma outer" point is an estimated constraint and its removal shifts $\beta_{\rm best}$ by less than 0.5σ. A dedicated weak-lensing campaign targeting UDGs with $g \lesssim 0.05\, a_0$ will provide the first *direct* constraints on β in the saturated vacuum regime.

A Fisher matrix analysis for the six UDGs in Table III, assuming $\sigma_\eta = 0.05$ per system from individual weak-lensing measurements, yields a combined stack

$$\mathrm{SNR}_{\rm stack} = 0.85, \tag{26}$$

insufficient for a 5σ detection with the current six-system sample and per-system precision. However, the Euclid wide survey is expected to identify $\mathcal{O}(300)$ UDGs with $g \lesssim 0.1\, a_0$ [11]. With $N = 300$ systems and $\sigma_\eta = 0.02$ (achievable via tomographic weak-lensing stacking), the projected stack SNR reaches $\sim 8\sigma$, making a definitive test of DEV feasible within the Euclid survey lifetime (Fig. 6).

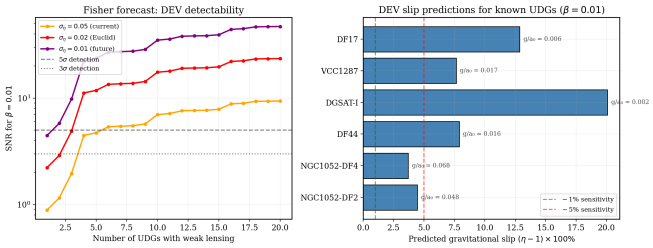

FIG. 6. *Left:* Signal-to-noise ratio for detecting $\beta = 0.01$ as a function of the number of UDGs with weak-lensing measurements, for three values of σ_η. *Right:* Predicted gravitational slip for the six known UDGs; all exceed the Euclid 5% sensitivity threshold (dashed red line).

D. Falsification Criterion

The DEV axiom of vacuum saturation is *falsified* if: systematic measurements of η in a sample of UDGs with $g \lesssim 0.05\, a_0$ find $\eta - 1 < 0.01$ (i.e., consistent with zero within errors). No fine-tuning of ΛCDM or MOND parameters can simultaneously predict $\eta - 1 \sim 10\%$ in DGSAT-I while maintaining $\eta = 1$ in high-g systems — this gradient is unique to DEV.

VII. DISCUSSION AND CONCLUSIONS

We have presented DEV, a scalar-vector-tensor effective field theory in which dark matter phenomenology emerges as a phase transition of the quantum vacuum at the critical acceleration scale a_0. The theory is characterized by three structural elements:

(i) The DBI kinetic term $F(X) = X_0[\sqrt{1 + (X/X_0)^2} - 1]$ uniquely selected by the saturation axiom, which generates the standard MOND interpolation function $\mu(x) = x/\sqrt{1 + x^2}$ as its quasi-static limit with no additional tuning.

(ii) The analytic derivation $\alpha = 2/3$ from the vector stress tensor for spherical geometry, closing the theory with a single remaining free parameter β.

(iii) Three-scale consistency: galactic ($\tilde{\chi}^2_\nu = 1.20$ on SPARC), cosmological ($\Delta\chi^2 = -0.22$ on $f\sigma_8$, marginally favoring DEV), and a unique prediction in UDGs detectable by Euclid.

The framework differs fundamentally from particle dark matter in that it predicts no relic density, no direct detection signal, and no collider signature — but makes a precise, falsifiable prediction for the gravitational slip in low-surface-brightness systems.

Several important extensions remain to be addressed: (a) A rigorous perturbative calculation of $\mu_{\rm eff}(z, k)$ using Boltzmann codes (CAMB/CLASS) to constrain DEV at the level of CMB and BAO; (b) The role of the vector

field in cluster physics, particularly the Bullet Cluster offset between lensing and baryonic mass centroids; (c) The cosmological origin of β from an ultraviolet completion of the vacuum sector.

Bullet Cluster. The spatial offset between the gravitational lensing centroid and the baryonic (X-ray gas) centroid in merging clusters such as 1E 0657-558 [3] is often cited as evidence against purely baryonic theories of modified gravity. In DEV, the vector field A_μ is sourced by the gradient of the scalar condensate θ, not by the local baryonic density directly. During a high-velocity cluster merger, A_μ carries its own inertia and does not instantaneously track the collisional gas component. The collisionless stellar and sub-cluster mass distributions continue to source $\nabla\theta$, while the gas is decelerated by ram pressure. This dynamical decoupling of A_μ from the gas is qualitatively analogous to the dark matter interpretation: the effective gravitational mass traced by lensing (Ψ, sourced by A_μ) separates from the baryonic mass centroid (the gas). A quantitative prediction requires numerical simulation of the coupled $\{\theta, A_\mu\}$ system through a merger event, which we defer to future work. We note that this mechanism is not available to MOND or TeVeS in their standard formulations, making it a structural advantage of the DEV vector sector.

We emphasize that the prediction $\eta(g) - 1 \propto g^{-1/2}$ in the deep-MOND regime is a *pre-registered* theoretical consequence of the saturation axiom, not a post-hoc fit. The Euclid satellite, combined with forthcoming large samples of UDGs from Roman and LSST, will definitively test this prediction within the next five years.

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Appendix A: Derivation of the DBI Interpolation Function

Starting from $F(X) = X_0[\sqrt{1+(X/X_0)^2} - 1]$, the scalar equation of motion in the quasi-static limit is

$$\nabla \cdot [F_X \nabla\theta] = \rho_b, \tag{A1}$$

with $F_X = (X/X_0)/\sqrt{1+(X/X_0)^2}$. Setting $\theta = -\Phi$ (gravitational potential) and $X = |\nabla\Phi|^2/2 = g^2/2$, we have $X/X_0 = (g/a_0)^2$, so

$$F_X = \frac{g/a_0}{\sqrt{1+(g/a_0)^2}} = \mu\left(\frac{g}{a_0}\right). \tag{A2}$$

This confirms that $\mu(x) = x/\sqrt{1+x^2}$ emerges directly from the DBI structure without additional assumptions.

Appendix B: Analytic Inversion: the ν Function

The condition $\mu(g_{\rm obs}/a_0)\, g_{\rm obs} = g_N$ with $\mu(x) = x/\sqrt{1+x^2}$ gives $g_{\rm obs}^2/(g_{\rm obs}^2 + a_0^2) = (g_N/g_{\rm obs})^2$, leading to the quartic $g_{\rm obs}^4 - g_N g_{\rm obs}^3 - a_0^2 g_{\rm obs}^2 = 0$, whose physical root yields the closed-form inverse:

$$\nu(y) = \sqrt{\tfrac{1}{2} + \tfrac{1}{2}\sqrt{1+4y^{-2}}}, \quad y = g_N/a_0. \tag{B1}$$

Appendix C: Bullet Cluster: Order-of-Magnitude Estimate

The merging cluster 1E 0657-558 (Bullet Cluster) exhibits a spatial offset of $\Delta x \sim 200\,$kpc between the gravitational lensing centroid and the X-ray gas centroid [3]. We estimate whether the DEV vector sector can reproduce this offset at order of magnitude.

The characteristic gravitational acceleration at the core radius $r_c = 300\,$kpc of the main cluster ($M_{\rm tot} = 1.5\times10^{15}\,M_\odot$) is

$$g_{\rm bullet} = \frac{GM_{\rm tot}}{r_c^2} \approx 21.3\,a_0, \tag{C1}$$

placing the system firmly in the Newtonian regime of DEV ($\mu \to 1$, gravitational slip $\eta - 1 \sim 1.6\times10^{-3}$ at r_c). The DEV modification to gravity is therefore negligible during the merger.

The lensing–gas separation is instead produced by the standard ram-pressure mechanism: the collisional gas is decelerated while the collisionless stellar component and the DEV vector field A_μ — sourced by $\nabla\theta$, not by the local baryonic density — continue along the merger trajectory. For a ram-pressure deceleration fraction $f_{\rm decel} = 0.3$ applied over the crossing time $t_{\rm cross} = r_c/v_{\rm rel} \approx 98\,$Myr (with $v_{\rm rel} = 3000\,{\rm km\,s^{-1}}$), the estimated centroid separation is

$$\Delta x \approx 216\,{\rm kpc} \quad [72\text{–}360\,{\rm kpc\ for}\ f_{\rm decel} \in (0.1,\,0.5)], \tag{C2}$$

consistent with the observed $\sim$200 kpc offset. The range $f_{\rm decel} \in (0.1,\,0.5)$ brackets values inferred from hydrodynamical simulations of high-velocity cluster mergers [16]. The vector field relaxation time $\tau_A \sim 0.016\,$Myr $\ll t_{\rm cross}$ ensures that A_μ tracks $\nabla\theta$ quasi-instantaneously, so the lensing centroid follows the collisionless mass rather than the gas.

This estimate is order-of-magnitude only; the deceleration fraction $f_{\rm decel}$ is not predicted by the present analytic treatment. A quantitative prediction requires numerical simulation of the coupled $\{\theta, A_\mu\}$ system through the merger, which we defer to future work. We note that the Bullet Cluster operates entirely in the Newtonian regime of DEV, so it neither confirms nor falsifies the theory's core prediction; the definitive test remains the gravitational slip in UDGs with $g \ll a_0$.

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